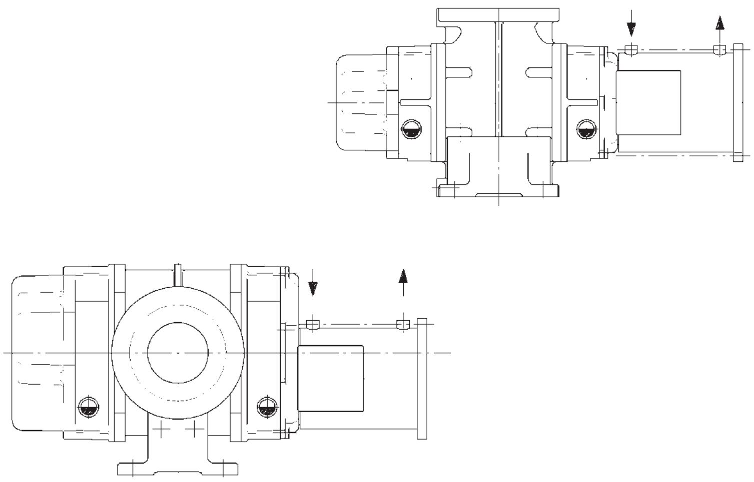
*Vacuum*

# Rotary piston blower stage

## Water-cooled canned drive

ENGLISH

**Installation, assembly and operating instructions**



AERZEN

AERZENER MASCHINENFABRIK GMBH

| G4-014 K EN | |
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| 180 184 000 | 12-2012 |


Die INFO-Seite ist vor der Inbetriebnahme durchzulesen.
Dort evtl. vermerkte Hinweise und Änderungen sind durchzuführen.

Read the INFORMATION sheet prior to commissioning.
Possible notes and changes indicated herein are to be effected.

La page INFO est à lire avant la mise en route.
Y apporter éventuellement des annotations et modifications.

De INFO-Bladzijde moet voor de inbedrijfname worden doorgelezen.
Daar eventueel opgeschreven aanwijzingen en modificaties moeten worden uitgevoerd.

Prima della messa in esercizio leggere la pagina INFO, ed eseguire eventuali
istruzioni o modifiche indicate.

Antes de proceder a la puesta en marcha, leer detenidamente la página informativa y
cumplir eventuales indicaciones y modificaciones indicadas en la misma.

A página de informações deve ser lida antes da colocação em funcionamento. As eventuais indicações e alterações
aí mencionadas devem ser respeitadas.

INFO-siden skal læses igennem inden idriftsættelsen. Evt. anvisninger og ændringer der står dér skal gennemføres.

Les INFO-siden før igangsetting. Anvisninger og endringer som står oppført der, skal utføres.

Läs igenom INFO-sidan före idrifttagning. Eventuellt angivna anvisningar eller förändringar skall genomföras.

Pidätämme oikeuden painovirheisiin, erehdyksiin sekä teknisiin muutoksiin..

***Druckfehler, Irrtümer sowie technische Änderungen sind vorbehalten.***

***We are not liable for misprints, errors, and we reserve the right to make technical changes.***

***Sous réserve de fautes d'impression, d'erreurs et de modifications techniques.***

***Drukfouten, vergissingen en technische wijzigingen voorbehouden.***

***Salvo error u omisión. Reservado el derecho a realizar modificaciones técnicas.***

***Salvo errori di stampa, incorrettezze e modifiche tecniche.***

***Reservamo-nos o direito a erros de impressão, enganos e alterações técnicas.***

***Der tages forbehold for trykfejl, fejl og tekniske ændringer.***

***Med forbehold om trykkfeil, feiltagelser og tekniske endringer.***

***Tryckfel, övriga fel samt tekniska förändringar förbehålles.***

***INFO-sivu on luettava ennen käyttöönottoa. Siellä ilmoitetut mahdolliset muutokset tai lisäykset on otettava huomioon***

# Inhalt / Contents




Aerzener Maschinenfabrik
Abteilung Technische Dokumentation
Technische Illustration & Redaktion
H.Nickel Technischer Autor

# Konformitätserklärung für Maschinen gemäß Maschinenrichtlinie 2006/42/EG

**Declaration of conformity** for machines in accordance with Machine Directive 2006/42/EC

**Déclaration de conformité** pour machines selon la directive Machines 2006/42/CE

**Die Konformitätserklärung für diese Drehkolbenmaschine wird von den technischen Angaben im Kapitel „Leistungsdaten“ ergänzt. Die dort erwähnten Angaben identifizieren das Produkt und sind in Verbindung mit dieser Konformitätserklärung zu verwenden.**
**The declaration of conformity for the rotary piston machine supplimented by the technical specification in the chapter „Performance data“. The data mentioned there identify the product and must be used in association with this declaration of conformity.**
**La déclaration de conformité de cette machine à pistons rotatifs est complétée par les indications techniques du chapitre « Caractéristiques de puissance ». Les indications qui y sont données identifient le produit et sont à mettre en relation avec la présente déclaration de conformité.**

*DEUTSCH* *Originalkonformitätserklärung*
Hiermit erklärt der Hersteller: **Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
dass die Maschine konform ist mit den Bestimmungen der folgenden Richtlinien:
Maschinenrichtlinie 2006/42/EG
Elektromagnetische Verträglichkeit 2004/108/EG.
Druckgeräterichtlinie 97/23/EG
Die Schutzziele der Niederspannungsrichtlinie wurden gemäß Anhang I, Nr. 1.5.1 der Maschinenrichtlinie eingehalten.
Die Konformitätserklärung bezieht sich auf den vom Hersteller in Verkehr gebrachten originalen Maschinenzustand.
Bei nachträglich durchgeführten Veränderungen und/oder nachträglich vorgenommenen Eingriffen erlischt diese Konformitätserklärung.
Die zur Maschine gehörenden speziellen technischen Unterlagen nach Anhang VII Teil A wurden erstellt.
Dokumentationsverantwortlicher: Herr Irtel Aerzen, 28-06-2011

*ENGLISH* *Translation of the original declaration of conformity*
The manufacturer: **Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
hereby declares that the machine conforms to the provisions of the following directives:
Machine Directive 2006/42/EC.
Electromagnetic Compatibility 2004/108/EC.
Pressure Equipment Directive 97/23 EC
The safety objectives of the Low Voltage Directive have been complied with according to Appendix I, No. 1.5.1 of the Machine Directive.
The declaration of conformity refers to the machine in its original state as marketed by the manufacturer.
Any changes introduced subsequently and/or interventions carried out subsequently will void this declaration of conformity.
The special technical documents for the machine in accordance with Appendix VII Part A have been created.
Responsible for documentation: Mr. Irtel Aerzen, 28-06-2011

*FRANÇAIS* *Traduction de la déclaration de conformité d'origine*
Le fabricant : **Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
déclare par la présente que la machine est conforme aux dispositions des directives suivantes :
Directive Machines 2006/42/CE
Compatibilité électromagnétique 2004/108/CE
Directive Équipements sous pression 97/23/CE
Les objectifs de protection de la directive Basse tension ont été respectés conformément à l'annexe I, n° 1.5.1 de la directive Machines.
La présente déclaration de conformité se rapporte à l'état d'origine de la machine tel que mis en circulation par le fabricant.
Cette déclaration de conformité est rendue caduque par toute modification et/ou intervention ultérieure.
La documentation technique pertinente pour la quasi-machine a été constituée conformément à l'annexe VII partie A.
Responsable de la documentation : Monsieur Irtel Aerzen, 28-06-2011

(Herr Irtel)
Leiter Techn. Abteilung
Head of the dept.
Directeur technique

Unterschrift des Herstellers
Signature of the manufacturer
Signature du constructeur

**A3-050 F XT**
1 von 2 06-2011

**Diese Konformitätserklärung ist mit dem Inkrafttreten der Maschinenrichtlinie 2006/42/EG ab dem 29-12-2009 gültig.**
This declaration of conformity applies when the Machine Directive 2006/42/EC comes into force on 29-12-2009.
Cette déclaration de conformité est valable à partir du 29/12/2009 avec l'entrée en vigueur de la directive Machines 2006/42/CE.

AERZEN

**Conformiteitsverklaring** voor machines conform machinerichtlijn 2006/42/EG

**Declaración de conformidad** para máquinas, de conformidad con la Directiva de Máquinas 2006/42/CE

**Dichiarazione di conformità** per macchine secondo la Direttiva macchine 2006/42/CE

CE

**De conformiteitsverklaring voor deze roterende zuigercompressor wordt aangevuld door de technische informatie in het hoofdstuk „Specificaties“.
Aan de hand van de daar vermelde informatie wordt het product geïdentificeerd en de informatie moet in combinatie met deze conformiteitsverklaring toegepast worden.
La declaración de conformidad para esta máquina de émbolos giratorios es complementada por los datos técnicos que se encuentran en el capítulo “Datos de potencia”.
Los datos ahí indicados identifican al producto y se deberán utilizar conjuntamente con esta declaración de conformidad.
La Dichiarazione di conformità per la presente macchina a pistoni rotanti è integrata dai dati tecnici riportati nel capitolo „Dati di prestazione“
Le informazioni incluse in quel capitolo identificano il prodotto e devono essere utilizzate insieme alla presente Dichiarazione di conformità.**

*NEDERLANDS* *Vertaling van de originele conformiteitsverklaring*

Hierbij verklaart de fabrikant: **Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
dat de machine voldoet aan de bepalingen van de volgende richtlijnen:
machinerichtlijn 2006/42/EG
elektromagnetische compatibiliteit 2004/108/EG.
richtlijn drukapparatuur 97/23/EG
De doelstellingen voor veiligheid van de laagspanningsrichtlijn zijn conform bijlage I, nr. 1.5.1 van de machinerichtlijn nageleefd.
De conformiteitsverklaring heeft betrekking op de originele machinetoestand die door de fabrikant in omloop is gebracht.
Bij veranderingen en/of ingrepen die naderhand uitgevoerd zijn, komt deze conformiteitsverklaring te vervallen.
De speciale, bij de machine horende, technische documentatie conform bijlage VII deel A is opgesteld.
Verantwoordelijk voor de documentatie: de heer Irtel Aerzen, 28-06-2011

*ESPAÑOL* *Traducción de la declaración de conformidad original*

Por la presente, el fabricante: **Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
declara, que la máquina cumple con las disposiciones de las siguientes directivas:
Directiva de Máquinas 2006/42/CE
Compatibilidad Electromagnética 2004/108/CE
Directiva de Equipos a Presión 97/23/CE
Los objetivos en materia de seguridad establecidos en la Directiva de Bajo Voltaje se han aplicado según el anexo I, núm. 1.5.1 de la Directiva de Máquinas.
La declaración de conformidad se refiere al estado original de la máquina que el fabricante comercializa.
En el caso de realizar modificaciones y/u operaciones posteriores, la declaración perderá toda validez.
La documentación técnica correspondiente a la máquina, de conformidad con el anexo VII, parte A ha sido elaborada.
Responsable de la documentación: Sr. Irtel Aerzen, 28-06-2011

*ITALIANO* *Traduzione della dichiarazione di conformità originale*

Con la presente dichiarazione il produttore:**Aerzener Maschinenfabrik GmbH, Reherweg 28, D-31855 Aerzen**
dichiara che la macchina è conforme alle disposizioni delle seguenti direttive:
Direttiva macchine 2006/42/CE
Compatibilità elettromagnetica 2004/108/CE
Direttiva sulle attrezzature a pressione (PED) 97/23/CE
Gli obiettivi di sicurezza della Direttiva bassa tensione sono stati rispettati secondo l’allegato I, n. 1.5.1 della Direttiva macchine.
La dichiarazione di conformità si riferisce allo stato della macchina originale messa in circolazione dal produttore.
In caso di modifiche eseguite successivamente e/o di interventi eseguiti in un secondo momento, decade l’obbligo della dichiarazione.
È stata predisposta la documentazione tecnica speciale relativa alla macchina secondo l’Allegato VII Parte A.
Responsabile della documentazione: Sig. Irtel Aerzen, 28-06-2011

(Herr Irtel)

Hoofd technische afdeling
Director Dpto. Técnico
Responsabile reparto tecnico

Handtekening van de fabrikant
Firma del fabricante
Firma del fornitore

Deze conformiteitsverklaring is geldig bij de inwerkingtreding van de machinerichtlijn 2006/42/EG vanaf 29-12-2009.
Esta declaración de conformidad obtendrá validez con la entrada en vigor de la Directiva de Máquinas 2006/42/CE, a partir del 29.12.2009.
Con l’entrata in vigore della Direttiva macchine 2006/42/CE la presente Dichiarazione di conformità è valida a partire dal 29-12-2009.

## Ersatzteile, *spare parts, pièces dé tachée, onderdelen, repuestos, pezzi di ricambio*

**- AERZENER MASCHINENFABRIK -**

**Ersatz- und Zubehörteile**
Es wird darauf hingewiesen, dass nicht von uns gelieferte Originalteile und Zubehör auch nicht von uns geprüft und freigegeben sind. Der Einbau oder Anbau sowie die Verwendung solcher Produkte kann daher unter Umständen konstruktive vorgegebene Eigenschaften der Anlagen beeinflussen. Für Schäden, die durch Verwendung von nicht Originalteilen und Zubehör entstehen, ist jede Haftung des Herstellers ausgeschlossen.

**Spare parts and accessories**
We draw your attention to the fact that original parts and accessories not supplied by us are also not inspected and released by us. Therefore, the installation and application of such products might influence under certain circumstances constructively stipulated properties of the plants. Consequential damages due to application of non-original parts and accessories release the manufacturer from any warranty and liability.

**Accessoires et pièces de rechange**
Nous attirons votre attention sur le fait que les accessoires et pièces d'origine n'étant pas de notre fourniture ne peuvent être controlés et pris en considératoin lors d'une réclamation. L'intégration ou le montage ainsi que l'utilisation de telles pièces peut influencer sous certaines conditions les caractéristiques et performances de la machine. Pour tout dommage causé du fait de pièces n'étant pas d'origine ou de montage erroné, nous déclinons toute responsabilité.

**Reservedelen en toebehoren**
Er wordt uitdrukkelijk op gewezen dat niet door ons geleverde originele delen en toebehoren ook niet door ons getest en vrijgegeven zijn. De in of aanbouw alsmede de toepassing van zulke producten kan derhalve onder zekere omstandigheden constructief gegeven eigenschappen van de installatie beïnvloeden. Voor schaden, die door gebruik van niet originele delen en accessoires ontstaan, is iedere aansprakelijkheid jegens de fabrikant uitgesloten.

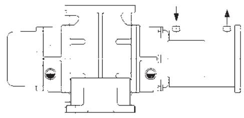
**Ricambi ed accessori**
Devono essere utilizzati solo parti e ricambi originali in quanto verificati dal costruttore della macchina. Il montaggio o l'impiego di prodotti non originali può, in certe circostanze, provocare un cattivo funzionamento dell'impianto. Danni causati dall'impiego di parti e/o ricambi non originali esonerano il produttore da ogni responsabilità e garanzia.

**Piezas de repuesto y accesorios**
Indicamos expresamente, que aquellos repuestos y/o accesorios no suministrados por nosotros no están comprobados ni homologados por Aerzen. Su montaje, así como su utilización pueden tener incidencia en las características prefijadas de la instalación. Por lo tanto no asumimos garantía ni responsabilidad alguna sobre éstas piezas y de los eventuales daños posteriores y/o alteraciones de las calidades y prestaciones de origen. Para daños originados por la utilización de piezas y accesorios no originales, se excluye cualquier responsabilidad por parte del fabricante.

## Aerzener Maschinenfabrik

# Leistungsdaten

*Performance data*



# Contents

## Suitability/General information

The German version of these operating instructions represents the „original operating instructions".
Versions in all other languages are a „translation of the original operating instructions".

Aerzener rotary piston machines are suitable for the oil-free conveying of air and neutral gases, as well as for use in special applications in industrial high vacuum technology.

Note / synthetic resin can --> In the case of the transport of aggressive gases, trials on resistance are to be undertaken prior to operation. These trials are to be validated and recorded. If a lack of suitability is found, the SRM series is to be used.

The technical performance limits must be observed if perfect operation of the equipment is to be ensured in the long term.
The performance limits stated in the order confirmation apply.
On this topic, please also note the blower-specific K0 and the charts on safe operation, as well as the tables in the appendix. The related charts can be requested from the manufacturer.

The intake temperature $t_1$ specified on the order confirmation serves as the installation site ambient temperature.

Non-observance of the technical performance limits or the safety information will absolve Aerzener Maschinenfabrik of its warranty obligations and its liability to pay damages or compensation as a result of any consequential damage. The same applies to defects that can be traced back to recommended inspections not being performed correctly or on time.

The vibration behaviour of blowers and compressors is determined by the balance quality of the pistons / rotors etc.
At Aerzener Maschinenfabrik, the piston/rotor drive shafts are balanced using the „half-key balancing" technique.

The maintenance intervals according to these operating instructions must be followed and carried out properly. Complying with the maintenance specifications will ensure that the machine retains its value and will also promote operational reliability.

**Example of type identification:**

**GM / GL 2800 HM-N**

| | | |
|---|---|---|
| GM | = | Blowers vertical conveying |
| GL | = | Blowers horizontal conveying |
| 2800 | = | Volumetric flow rate at 50 Hz |
| HM | = | Hermetic motor |
| N | = | N > Mains frequency |
| | | C > High frequency, continuous mode |
| | | Y > High frequency, cycle mode |

ENGLISH

## Construction, mode of operation

**Blower stage:**
The blower stage is supplied as a fully assembled unit.
The following additional work must be carried out:
- Correct installation/assembly in the relevant unit/system
- Connect the conveying ducts.
- Check lubricating oil level.
- Top up lubricating oil level if necessary.
- Install any special accessories which have been supplied separately.

**Motor connection:**
Electrical installation of the drive motor must only be carried out by an authorised electrical fitter.
Connect the motor and control voltage to a stable common network to ensure that the power contactor is no longer latched if the power supply fails. Voltage fluctuations and voltage dips must be avoided.
Alternative: An electronic monitoring relay must be installed in parallel to the drive motor so that the power contactor is no longer latched in the event of a power failure.
It must only be possible to restart the machine once it has stopped.

<u>Prerequisites for operating rotary piston machines with asynchronous electric motors on a 3-phase AC supply network:</u>
The machine should only be used on stable three-phase supply networks.
The voltage and frequency limits must be observed. These are specified in EN 60034-1.
Voltage fluctuations/dips beyond the tolerance zone can lead to serious damage to all the drive system elements, such as couplings, V-belts, V-belt pulleys, shafts and gear wheels, etc. .
If excessive voltage fluctuations arise in the system,
Aerzener Maschinenfabrik recommends the following measures in order to prevent the blower, compressor or motor from sustaining damage:
- Use suitable protective equipment that will shut down the motor and safeguard it against an automatic restart when inadmissible operating data are detected. Also comply with EN 60034-1 and EN 60 204-1 in this respect.

**Unit/System:**
The blower stage must be installed/assembled in a unit/system in the correct way so that it is safe to operate.
There must be no forces or moments acting on the flange that could distort the blower housing.
The assembly/installation must be guaranteed to be resistant to distortion.
The blower stage must be installed level and flat.
Operational safety/reliability will be affected to the extent of any misalignment.

**Startup screen:**
It must be ensured that the intake gas is free of dirt and impurities to ensure the the blower stages are operating reliably.
If the intake is from a duct, we recommend using a startup screen.
The degree of contamination or the specifications on the maintenance schedule determines when the startup screen must be cleaned or replaced.

**Mode of operation:**
Aerzener rotary piston blowers are dual-shaft rotary piston machines with pistons which rotate uniformly in opposite directions. Timing gear wheels ensure that the rotary pistons run contact-free.
The direction of rotation defines the conveying direction of the blower.
At the operating point, the medium to be conveyed flows through the intake flange into the housing. From there, it is forced through the conveying chambers (consisting of the pistons and the blower cylinder) in the direction of the outlet.
The installation is driven by a canned motor which is hemetically flanged in front of the drive shaft.

The conveying chamber (cylinder) is sealed off from the oil chambers (housing cover and gear case) by a combined oil retainer partition and rectangular ring labyrinth seal.

This type of sealing is not hermetically sealed. In some instances, special measures might be necessary for short pump cycles, in order to prevent oil flowing into the conveying chamber.
Please contact Aerzener Maschinenfabrik GmbH for further information on this topic.

The level of the lubricating oil for supplying the bearings and the gear wheels can be seen in the oil sight glasses. It is essential to keep an eye on the MAX. level.
If the oil level is too high, oil can penetrate into the conveying chamber in an uncontrolled manner.

When gases are conveyed, compression heat is produced. Some of the heat is dissipated via the outer surfaces of the blower and conveying duct into the ambient air. Outer surfaces and conveying ducts reach temperatures which can cause burns to exposed skin.

**Foreseeable incorrect operation or use**
Dust contaminants should be removed before entry into the blower. In particular, substances which can settle in the conveying chamber or on the rotors can jeopardise the operational reliability of the machine.
The same applies to the compression of resublimating gases containing solid particles, which can separate from the gas phase and deposit in the machine. Resublimation in the blower must be prevented by using suitable process control measures (pressure, partial pressure, temperature and speed etc.).

Exceeding the performance limits may lead to overheating in the rotary blower and jeopardise operational reliability. The pump-stand builder must run a test to check whether the designed pump process can cause the blower to overheat. If necessary, special measures to ensure reliable use of the blower must be agreed with Aerzener Maschinenfabrik. It is particularly important to make sure that performance specifications supplied by Aerzener Maschinenfabrik relate to the delivery medium “air” and the lubricating oil “mineral oil”.

## Due care must be taken before and during operation

**CAUTION!** Identifies all hazardous situations

**WARNING !** Indicates direct risk to personnel.

**Erst lesen - dann bedienen!**

***Read first, then operate !***

Upon arrival/receipt, the rotary piston machine must be checked against the delivery and order documentation to make sure that it has not been damaged during transit and is complete.
Work safety regulations, safety information and operating instructions must all be complied with.

Read the **INFORMATION sheet** prior to commissioning. Any notes or changes stated on it must be observed.

The tasks described below must only be carried out by qualified personnel who are familiar with the purpose and operation of the machine and its components, and have been instructed in the relevant safety information.

This rotary piston machine conforms to European safety regulations. Nevertheless, residual technical risks, which could endanger persons and property, may remain. To prevent this, operators must take note of the following **safety information**:

- **Assembling the blower stage in the unit**
  Keep the flange connections sealed until they are about to be installed.
  The ingress of dirt, burring, sputtering, liquids and similar must be avoided. Otherwise the blower stage may become blocked or sustain serious damage.

**Information about operating data**

- The operating data must be observed.
  The machine must be used appropriately and in compliance with the regulations, and within its performance limits.
- The sound pressure level may deviate from that stated in the operating data, depending on the operating state. A sound pressure level in excess of 85 dB(A) may therefore occur for a short period of time.

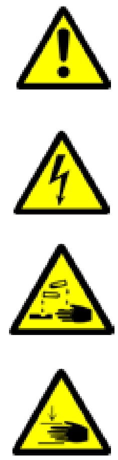
**Information about general operation**

- The general safety and industrial accident prevention regulations set down by law must be taken into account in addition to the information in the operating instructions.
- The user is responsible for making sure that the machine is only run if it is in perfect working order, safe to operate and in its original condition.
  Damaged compressors or compressors which are not in perfect working order must be replaced immediately.
- Any operation that impairs machine safety is prohibited.
- The supplier documentation provided with the accessories must also be observed.
- Protective equipment such as the belt guard / coupling guard, hood elements, electrical safety elements, valves, motor protection/EMERGENCY OFF etc., must not be removed or its operational safety restricted whilst the machine is in operation. **Risk of injury!**
- Do not operate the machine if electrical, mechanical or hydraulic connections are defective or missing.
- It is expressly prohibited to operate the rotary piston machine without appropriate protective equipment or safety devices.
- The machine must only be used in stable three-phase supply networks. Voltage fluctuations/dips outside the tolerance zone can seriously damage all the elements that make up the drive system. These include couplings, V-belts, V-belt pulleys, shafts, etc.

- **CAUTION!** Mechanical damage will result if the machine is operated without sufficient oil!
- The safety, operating and maintenance information of the drive motor manufacturer must be followed!
- Follow the instructions of the accessories manufacturer, as well as the general safety regulations.
- If the conveying medium is helium or oxygen etc. the „Safety data sheets“ and the general safety instructions as well as the related safety regulations must be followed!

**Safety instructions for commissioning**

- You must read and understand these operating instructions before commissioning the machine.
- Commissioning must only be carried out by persons with the relevant knowledge and skills.
- Before switching on the machine, ensure that you are familiar with all protection, operation and monitoring elements by referring to these instructions.
- Prior to first use, check that the intake side is clean.
  Any evidence of dirt, dust or foreign matter must be removed from the intake area.
- An adequate supply of lubricating oil must be ensured during operation.
- Compliance with the maintenance schedule is absolutely mandatory.
- The operator must use oil of a suitable quality, conforming to the Aerzener lubricating oil specifications.
- It must be ensured that a stopped positive displacement machine cannot be moved from the rest position without operating the start function, no matter what the cause!
- Damaged rotary piston machines or machines that are not in perfect working order must be replaced immediately.



**Qualification of operators**

- Each person dealing with the installation, operation, maintenance and repair of this unit must have read and understood the operating instructions.
- The unit must only be operated by trained and authorised personnel.
- Personnel must undergo training with reference to the operating instructions.
- Responsibility for operation must be precisely defined in order to prevent any unclear designation of responsibilities.
- Operators must be proficient and instructed and appointed for the work.
- Work on current-carrying components must only be carried out by trained and authorised specialist personnel. The unit must be disconnected from the power supply. Fuses must be disconnected.

**Safety instructions in relation to prevailing residual risks**

- The warning and information signs on the machine must be observed.
  They provide important information about potential sources of danger.
- Before commissioning, check that the machine is not damaged in any way.
- Do not operate the unit if electrical connections are damaged, defective or not properly connected.
- Do not operate the unit with an open intake or discharge nozzle, as rotary piston machines are positive displacement machines which present a risk of injury in the area around the conveying chamber.
- Only suitable tools that correspond to the respective standards and design of the bolts, nuts and screwed connections must be used.
- When using cleaning agents and sprays, there is a **risk of poisoning** from inhalation, and a **risk of burning** from coming into contact with them.

**Warning signs for hazardous operations**

- Repairs and modifications to the unit must only be carried out in a professional manner. If you have any problems, please contact Aerzener After-Sales Service for assistance.
- Before carrying out any retrofitting, servicing or maintenance work that requires protective equipment to be removed, the power supply must be disconnected and the machine secured to prevent it starting up.

- All lubricating and control oil pipes should only be retightened or opened while the machine is in a depressurised state.
- Depressurise conveying ducts before attempting to remove them.
  Allow a sufficient delay of several minutes for the lines to depressurise.
- Screwed connections should only be retightened while the machine is in a depressurised state!

- Take note of the oil temperature when changing the oil. The oil temperature must not be above 60°C. If it is above 60°C, there is a
  **risk of burns**!

**Personal protective measures**

- Conveying ducts or components on the discharge side must not be touched without adequate protection. The ducts and components can reach temperatures in excess of 70°C.
  Risk of burns!
- There is a **risk of burns**
  Wear protective clothing.
- Close-fitting clothing is required due to the presence of rotating components.
  **Risk of injury!**
- **Use ear defenders** when the machine is operating!

**Information about the installation site**

- It is the operator's responsibility to use and operate the machine in accordance with its intended use taking into account the local conditions.
- The machine must only be operated in a suitable, well-ventilated installation site.
  The installation site must be free from excessive dust, acids, steam and explosive or flammable gases.
- The protective equipment provides protection against injury and must not be modified or bypassed.
- The machine is to be installed such that objects are safely prevented from
- falling on the machine!

**Instructions and responsibilities for taking care of the drive motor**

- All work on the motor must only be carried out when it is switched off and secured against switching on again.
- All work must only be carried out by qualified, responsible specialist personnel.
  Also comply with: VDE 0105, IEC 364.
- Low voltage machines may have dangerous, live, rotating components as well as hot surfaces.
- You must follow the safety instructions when using cleaning agents.
- It is only permitted to use the lifting eye on the motor housing to transport the motor as a separate component.
- Ensure that the machine has a proper supply of coolant.
- Fill the machine with lubricating oil according to its size.
- The cross section of the connection cable must be matched to the rated motor current according to the local regulations.
- Only tested cable glands may be used for the cable routing.
- Holes in the connecting terminal compartment which are „not" used must be closed with blind grommets.

**Instructions for running motors with liquid-cooled housings**

- The flow of coolant must only be limited by throttling the return line.
  This is the only way to guarantee that the cooling jacket is completely filled and the air bubbles are discharged.

# INADMISSIBLE OPERATION

- Installation on inclined and/or sloping foundations
- Failure to observe the operating data.
- Failure to observe the maintenance intervals
- Incorrect direction of rotation
- Switching on - while coasting down
  - when rotation is in reverse
- Inadmissible pressure increase
- Not reaching or exceeding the speed limit.
- Exceeding the limit discharge temperature / see also type plate.
- Changing poles at low speed before the motor stops
- Operating the machine without modules designed to ensure the safety of the personnel and machine or with such modules which are damaged or faulty.
- Overfilling beyond the max. oil level.
- Operating the machine without oil
- Exceeding the permitted motor currents. See Appendix I.
- Running with a insufficient supply of cooling water.
- Running without a supply of cooling water.

**“Reasonably foreseeable misuse” that can result from easily forseeable human behaviour:**

- Operating the machine without having filled the lubricant.
- Operating the machine with too much lubricant.
- Operating the machine with reduced intake performance, e.g. due to a soiled intake filter, starting strainer etc.
- Assembly and commissioning of the rotary piston machine with intake-side and/or discharge-side flange sealers, protection covers or similar.
- Insufficient ventilation at the installation site, no insulated lines.
- Operating the machine without an isolated protective device, open belt or coupling drive.
- Risk of injury from rotating components!

**Foreseeable incorrect operation or use**

- Dust contaminants should be removed before entry into the blower. In particular, substances which can settle in the conveying chamber or the rotors can jeopardise the operational reliability of the machine.
- The same applies to the compression of resublimating gases containing solid particles, which can separate from the gas phase and deposit in the machine. Resublimation in the blower must be prevented via suitable process control (pressure, partial pressure, temperature and speed etc.).

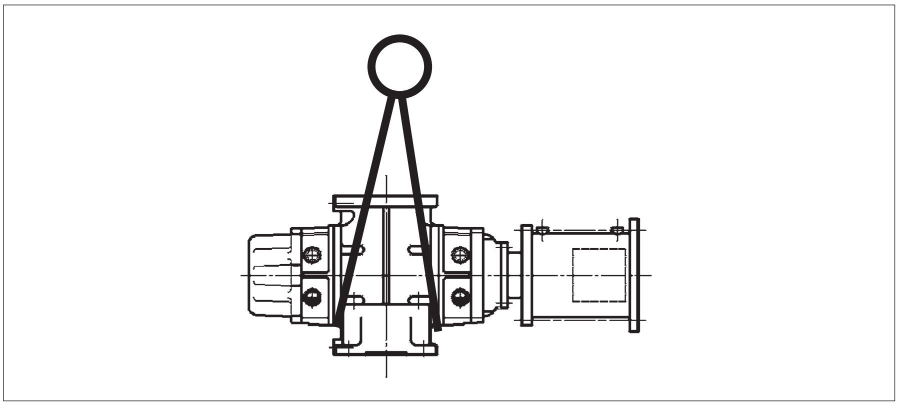
ENGLISH

## Transport/Installation

When transporting the machine, the following key points must be observed:

- Do not expose the blower to impact loads.
- Transport the blower using a crane, fork-lift truck, lift trolley or similar.
- Only suspend the blower as shown on the sketch.
- Use suitable hoisting gear that is appropriate for the load concerned.
  Failure to observe this point could result in damage or injury!
- The hoist, cables and chains etc. must be designed for the load concerned.
- The hoist must be arranged in accordance with the machine‘s centre of gravity.
- The hoist must not be able to exert any forces on the machine that could result in damage.





During installation the following points must be observed:

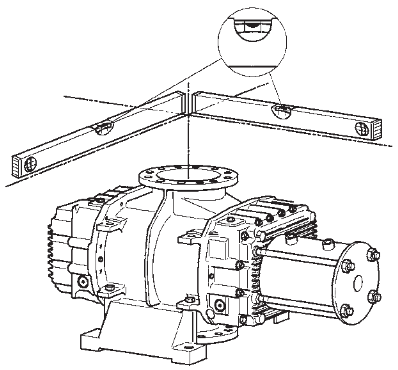
- Prior to delivery, rotary piston machines are treated and packaged to allow for a 12-month storage period. If they are to remain in storage following this 12-month period, they will need to be retreated; see storage and mothballing guidelines TN0 1175.
- While the units are in storage, ensure that they are correctly preserved and packaged and, if necessary, that they undergo nitrogen replenishment in accordance with TN0 1175. They must be checked every six weeks.
- You must eliminate any adverse influences immediately to ensure ongoing preservation of the units.
- Intermediate storage must be in dry, clean rooms that are free of vibrations.
- The oil chambers have been sufficiently preserved for one year.
- In the case of belt-driven machines, the V-belts should be slackened during storage.
- If the unit is intended to be taken out of operation for longer than six weeks, the conveying chamber, piston and bare parts must be preserved.
- The blower must be installed on a level, vibration-free and slope-free foundation and the connections made.
- CAUTION!  If the blower is installed with a tilt, this will result in irreparable damage to the machine caused by the uneven oil level.
  The machine‘s operational safety/reliability and service life will be affected in accordance with the extent of any misalignment.
- Remove all packaging material. If your machine features an acoustic hood, ensure that the inlet and exhaust air louvre profiles can be accessed.

- Ensure that the installation site is adequately ventilated.
- The following standard installation conditions must be maintained:

| | | | | |
|---|---|---|---|---|
| Ambient temperature | : | -10°C | to | 40°C |
| Rel. air humidity | : | 0% | to | 80% |

  Chemical-free atmosphere
- Check that the rotary piston machine can run freely. Sluggishness indicates the presence of distortion or foreign bodies.
- To prevent the units acquiring an electrostatic charge, the motor, acoustic hood and base frame must be earthed via the connections provided.
- Fasten the ducting on the discharge side and the ducting on the intake side separately to make sure the connection is stable.
- There must be no forces and/or moments acting on the flange or connections.
- Make sure that the blower runs freely before it is installed.
  Sluggishness indicates the presence of foreign matter inside the conveying chamber.
- Remember to remove the sealing covers from the intake-side and discharge-side flanges prior to installation!
- The blower must not be distorted by alignment or tightening operations.
- Make sure that the blower can run freely after it is installed.
  Sluggishness indicates the presence of foreign matter inside the delivery chamber or distortions.
- Take noise protection into consideration. Ducting and foundations may be induced to produce natural vibrations and the associated sound emissions.
- If you are planning the system in-house, the safety information, maintenance information and technical documentation provided by your component suppliers must also be followed.

**Before commissioning, check the oil level in the oil chambers on the blower stage and correct if necessary.**



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## Ducting

- When installing the ducting on the intake side, make sure that it is clean.
  Foreign matter and dirt will damage the blower.
- Any ducting which must be connected must be carefully cleaned and must be free of foreign matter.
- Ducting on the intake side must be supported.
  There must be no forces and/or moments acting on the blower housing.
- A start-up screen should be installed for the first 500 operating hours to prevent damage from contaminants.
  Install the start-up screen together with a seal on the intake side.
- Before tightening, the ducting and seals must be seated on the blower housing flange stress-free.
- The maximum distance in the sealing surface area is 0.05 mm on one side.
- Once it has been tightened, the blower stage should rotate freely without any resistance.
- Please note that ducting and foundations may be induced to producing vibrations and sound emissions when the blower is in operation.
- Therefore, insulation and soundproofing measures should be taken into consideration right from the planning stage.
- If you are planning a system in-house and/or plan to install the blower stage yourself, the safety information and technical documentation provided by your component suppliers must also be followed.

# Operating instructions and instructions for use

**Operating instructions and instructions for use for the vacuum blower with canned motor drive**

The maximum differential pressure of the blower is specified in Appendix II „Operating data“ in relation to the operating time.
The differential pressure specified in column 4 is only permissible for a duration of 5 minutes.
The blower must then be run for at least 15 minutes continuously at the differential pressure, Column 5 = cooling phase in the rated range.
This period can be shortened by reducing the voltage while operating within the rated range.
The permissible performance limits in Appendix II must be complied with,
The motors are fitted with up to 3 ptc thermistor detectors. To avoid damaging the motor, we recommend connecting the ptc thermistor detector to a monitoring control device.

Type of connection and voltages
The motors must be connected according to the data on the motor rating plate.

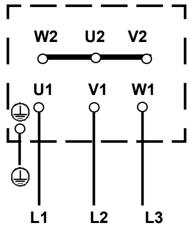
**A) Motors with the rating plate data**
**400V/50Hz and 460V/60Hz**
Can be run on a three-phase network without a frequency converter.
The motor windings are designed for Y-connection.
If the poles on the terminal board are connected in delta, the motors must be run at 265V/60Hz or 230V/50Hz.

***CAUTION!***
At low voltages, the motor load must be reduced to prevent it from overheating.

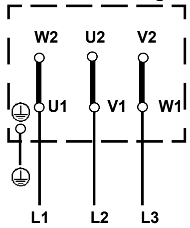
**Running without frequency converter at 460V/60Hz or 400V/50Hz (mains operation)**
In the connecting terminal compartment, 3 poles must be connected as the neutral. L1, L2 and L3 must be connected to the remaining poles.

**Running without frequency converter at 200-265V/60Hz or 200-230V/50Hz (mains operation)**
In the connecting terminal compartment, 6 poles must be delta connected to L1, L2 and L3.

**Sternschaltung**
***star switching***



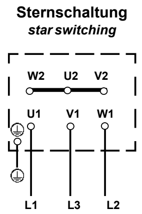

**Dreieckschaltung**
***delta switching***



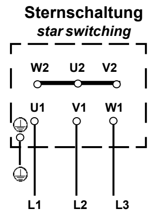
**B) Motors with frequency specifications not equal to 50 / 60 Hz**
may only be run with the frequency converter.
Running these motors on a three-phase network without a frequency converter is inadmissible.
The motor windings are designed for Y-connection.
The design voltage at the rated frequency is stipulated in Appendix I.

**Running at frequency converter design voltage**
**350-400V (line voltages L1, L2, L3)**
In the connecting terminal compartment, 3 poles must be connected as the neutral. L1, L2 and L3 must be connected to the remaining poles.

**GM 5500 / 7500 CM-C**

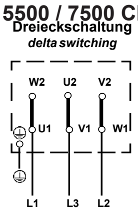



**remaining sizes**

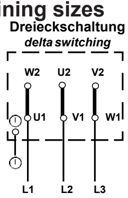



**Running at frequency converter design voltage**
**200-230V (line voltages L1, L2, L3)**
In the connecting terminal compartment, 6 poles must be delta connected to L1, L2 and L3.

**GM 5500 / 7500 CM-C**

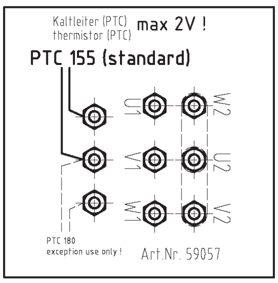


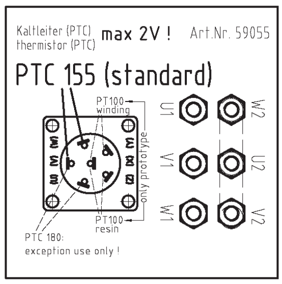

**remaining sizes**




**Connection / thermal winding protection**
All motors have a PTC sensor to monitor the winding temperature.

**Connection terminals / motor terminal strip**
2x2 pole thermistor connection terminals are provided on the motor terminal strip. Thus, the motor can be protected at 155°C or 180°C.
The PTCs must be connected to the frequency converter or to a suitable electronic selector in order to prevent damage to the motor due to overheating.
The PTC must be selected according to the operating data.
C- / N-operation: PTC 155
Y-operation : PTC 180
Also follow the connection instructions inside the motor terminal compartment.



**Operating instructions and instructions for use for the canned motor**

**Use**
Drive unit for vacuum blowers

**Operating modes**

**N** : Running at mains frequency 50 / 60 Hz on a three-phase supply network,
no frequency converter necessary.
**C** : Constant operation at max. speed, frequency converter mode possible,
no transient peak load possible.
**Y** : Periodic interval mode
Max. 5 minutes peak load; Min. 25 minutes partial load.

**Cooling**
Cooling medium : clean tapwater, pH = 7
Flow rate
of the cooling medium : see sep. Table: > ... Litres/minute

Operating pressure of the cooling medium : 2 bar ... 5 bar
Inlet temperature of the cooling medium : + 5°C to + 25°C

**Circuit diagrams**
The circuit diagrams for the motor connection are cast in the terminal box cover or listed in these operating instructions.

**Connection terminals / motor terminal strip**
2x2 pole thermistor connection terminals are provided on the motor terminal strip. Thus, the motor can be protected at 155°C or 180°C.
The PTCs must be connected to the frequency converter in order to prevent damage to the motor due to overheating.

**Direction of rotation**
The direction of rotation must be checked via the sight glass in the front of the motor cover.
Reverse the direction of rotation if necessary by swapping over the two outer pole lines.

***CAUTION!***
The wrong direction of rotation will damage the vacuum blower.

**Guarantee conditions**
As soon as the operator opens the drive motor or makes changes to the mechanical or electrical components, the guarantee is void. This does not apply to opening the connecting terminal compartment.
Proper maintenance as stipulated in the section „Maintenance“ does not count as intervention according to the guarantee conditions.
The guarantee agreed by the manufacturer is not thereby affected.

**Repair**
Only the manufacturer is allowed to carry out repairs and only the manufacturer is al-lowed to open the motor.

# Choice and requirements for the frequency converter

**Choice**

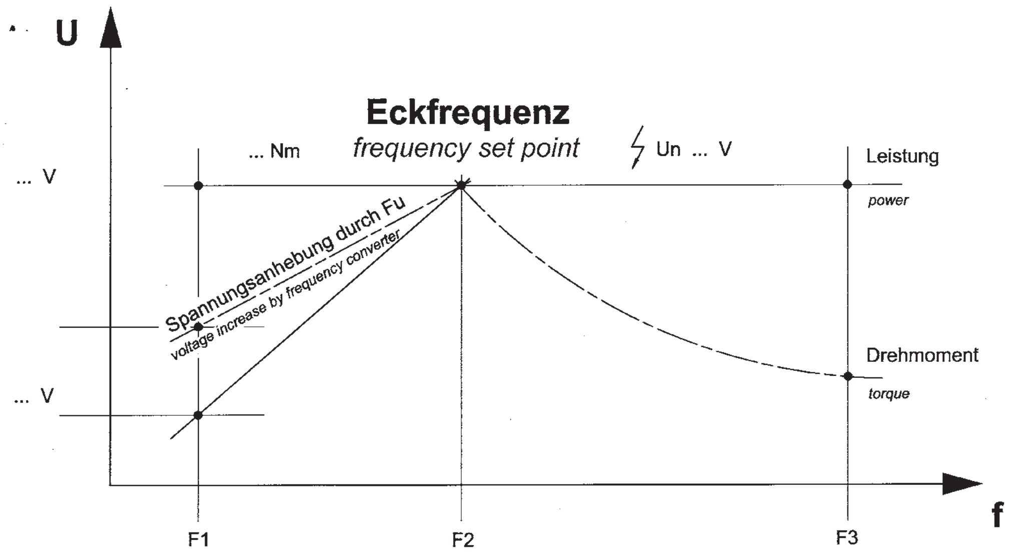
| | | |
|---|---|---|
| Output rated current | : | The rated current required for the motor is selected according to the converter rating plate. *Not according to the motor power!* |
| Mains voltage | : | According to the local conditions. The output voltage must not exceed the supply voltage! |
| Mains frequency | : | According to the local conditions |
| Output filter | : | Adapted to suit the converter |
| Cycle frequency (rated switching frequency) | : | > 15 kHz (select as high as possible) |
| Voltage dip | : | Also known as energy saver mode. Not used as a saver mode but as a component of the function. |
| Boost | : | Autoboost or adjustable. |

**Requirements**

All the parameters listed below must be adjusted and checked before commissioning. Parameters which are missing or incorrectly adjusted will destroy the drives!

| | | |
|---|---|---|
| Original operating instructions | : | Available, read through and understood |
| EMC principles | : | Installation checked |
| Operating voltage (mains) | : | According to the local conditions |
| | | Basic settings of the drive |

| | | |
|---|---|---|
| Frequency set point | : | According to the motor rating plate<br>At this adjusted frequency, the converter reaches its maximum output voltage.<br>If the frequency set point is not correctly adjusted, the motor may overheat! |




| | | |
|---|---|---|
| Rated current | : | According to the motor rating plate |
| Rated voltage | : | According to the motor rating plate |
| Rated frequency | : | According to the motor rating plate |
| Number of poles on motor | : | According to the motor rating plate |
| Cycle frequency | : | > 15 kHz |

Frequency converter parameter assignment principle
According to the frequency converter constant torque characteristic => the torque does not change with the speed!

| | | |
|---|---|---|
| Boost: | : | Autoboost or approx. 2% ... 5%<br>The voltage dip can be compensated for by the boost so that the pull-out torque of the motor remains almost constant over the entire voltage range. |

| | | |
|---|---|---|
| Voltage dip | : | Drops to a minimum of 70% of the motor rated voltage |
| (Energy saving mode) | | When the motor is operating in interval mode at reduced power, the rated voltage must be reduced to relieve the can. If the motor is run for longer than 10 minutes at the rated voltage (depending on the load), the motor will be destroyed! |
| Motor protection | : | PTC in the motor; connected to the converter! Earth installed! |
| Length of motor line | : | Shielded line; as short as possible |

**Commissioning**
The drive can be adapted to the requirements by programming the converter. Most manufacturers supply their converters with standard settings. It is essential that these are adjusted before commissioning the canned motor!

## Minimum requirements for the frequency converter

The maximum permissible rate of voltage rise for the frequency converter is 1200 V/ms. If this value is exceeded, e.g. by excessively long cables or the make of frequency converter etc. a motor choke / motor filter coil which matches the frequency converter must be fitted.
Omitting these components can damage the motor‘s insulation and lead to its breakdown.

**Voltage rise during the start-up phase**
The frequency converter has an auto-boost function.
The auto-boost function must be activated to improve the motor operating behaviour while running at low speed and when it is starting.
If no auto-boost function is available, the converter frequency characteristic can be increased by up to 10% of the design voltage at rated frequency.

**Voltage dip under partial load**
The frequency converter must have an automatic voltage dip device in the partial load range (energy saving mode).
Running without an energy saving mode can lead to serious damage to the motor.
This invalidates claims under the guarantee.
The voltage dip may be more than 50% of the design voltage at rated frequency, depending on the manufacturer.

1) GM 1000/1400 HM-Y, GM 2000/2800 HM-C:
max. n = 100 Hz possible without voltage dip
2) GM 2000/2800 HM-Y, GM 4000 HM-C:
max. n = 70 Hz possible without voltage dip
3) GM 4000 HM-Y, GM 5500/7500 HM-Y:
max. n = 50 Hz possible without voltage dip
4) GM 8000/9500 HM-C / Y:
no voltage dip necessary

The following frequency converters have been successfully tested:

| | | | |
|---|---|---|---|
| **Fuji** | **FRN-G11S** | | |
| **KEB** | **Combivert F5** | **/** | Not permissible with default, Aerzener parameter assignment. Take note of the Aerzener specifications in Appendix III. |

The size of the frequency converter is determined by the motor current required, pay attention to the technical data and Appendix I.

**Minimum parameter assignment for running with the frequency converter**
Design voltage at rated frequency 400V (Y-connection), 200-230V (Delta connection)
Frequency set point, see data on the motor rating plate and compare Appendix I.
To protect the motor, the maximum motor current must be parameterised according to the motor rating plate and Appendix I as the current limit. Particularly in networks where the motor is intended to be run up to the frequency set point at voltages below the design voltage at rated frequency, it is essential that the current limit be parameterised in order to protect the motor.
Starting at the current limit.
PTC monitor

**Electronic bypass regulator**
When using an Aerzener frequency converter, an „electronic bypass regulator“ is also available which protects both the Roots pump itself and any other pumps connected downstream from overload.
If there is an increase in the pressure on the intake side of the Roots pump, the electronic regulator adjusts the speed of the root pump accordingly.

Sudden increase in pressure => sudden reduction in speed.
(e.g. opening the door of a process chamber under vacuum)
If the permissible differential pressure load is exceeded, the Roots pump is suddenly stopped. After resting for 15 seconds, the Roots pump restarts and accelerates up to the maximum possible speed.

Gradual increase in pressure => gradual speed regulation.
If the intake pressures slowly increase or fluctuate, the frequency converter regulates the speed so that the maximum permissible differential pressure load is not exceeded.
From the point of view of the system, it is necessary to make sure that the Roots pump is only run outside the permissible pressure conditions briefly (5 minutes maximum).
Otherwise, the rotors will overheat and startup damage must be expected.
Reducing the speed appropriately protects the Roots pump and the backing pump from mechanical overload.
A mechanical bypass valve is not necessary.

# Commissioning

1. Install and assemble the blower stage in accordance with these instructions.

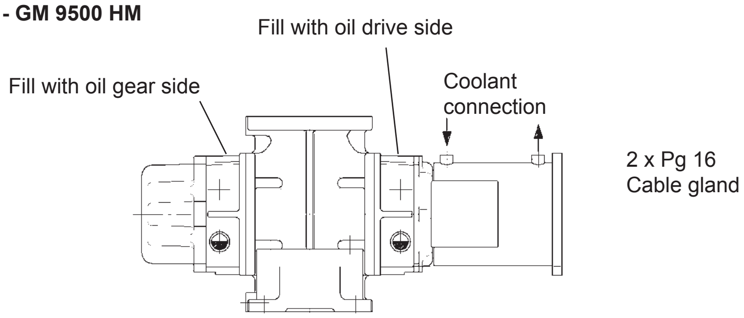
2. **When commissioning**
   Fill with lubricating oil.

   ***Blower*** : Fill with oil gear side and drive side

   Check lubricating oil level and correct if necessary.
   Check that the oil filler plug and the drain valve are seated securely and do not leak.

3. Connecting the coolant supply for the drive motor.
   Please take note of the following figure. The arrows indicate the permitted flow direction for the coolant.

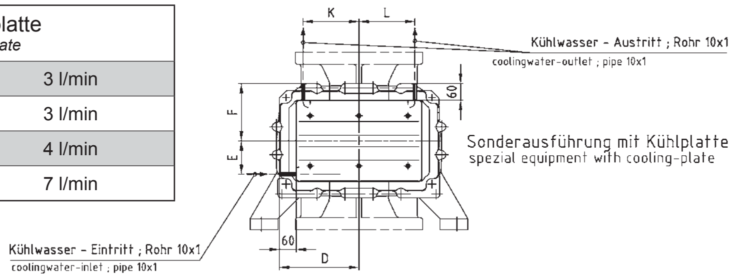
**GM 1000 HM - GM 9500 HM**

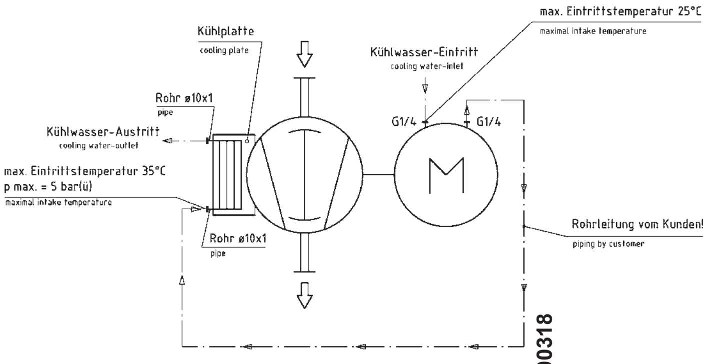



The coolant must be kept free of contaminants.
Provide a filter if there is a risk of contaminants capable of forming deposits.

Optional version
Connecting the coolant supply for cooling plate system.

| Kühlwasserbedarf Kühlplatte<br>*cooling water demand cooling plate* | |
|---|---|
| SRM/HDM 150/120 | 3 l/min |
| SRM/HDM 200/200 | 3 l/min |
| SRM/HDM 240/200 | 4 l/min |
| SRM/HDM 240/300 | 7 l/min |






PG-00318

4. Measuring the insulation resistance.
   Make the electrical connection.
   *Observe the motor*“right rotating field“
   With frequency-converter drive, adjust to *„right rotating field“*.
   Connect the PTC thermister protection / temperature alarm for the winding.

5. Operating data according to the performance curves.

6. Check direction of rotation. Note the direction of rotation data on the vacuum machine also.
   The motor must be started briefly for 1-2 seconds at most to check the direction of rotation.
   Do not run it for any longer under any circumstances since the wrong direction of rotation will damage the vacuum machine.
   ***CAUTION!*** The wrong direction of rotation will destroy the vacuum machine !

7. ***CAUTION!*** Do not switch on the machine while it is coasting down.
   If you do, there is a risk of causing serious damage !

8. Also comply with the information and operating instructions of the machine manufacturer.

9. Open the slide on the system side. The machine can now run.

10. Switch on the drive motor!
    Switch it off after approx. 20 seconds and monitor the blower to make sure it coasts down smoothly.
    Check the EMERGENCY OFF switch.

11. The unit is now ready for operation.

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## 9 Switch off / shutdown

- Switch off is effected via the power switch on the motor.
- To shut the unit down completely the fuses must be removed after the blower has come to a standstill. The valves on the conveying pipes must be closed. Avoid possible entry of condensate into the blower stage.
- In case of a shutdown for more than six weeks the conveying chamber is to be preserved and the blower turned regularly by hand to avoid damage due to stand-still. Observe also the TN01175 regulations governing storage and preservation.

In case of danger:
Press EMERGENCY - OFF button. For details refer to the instructions of the supplier of electrical components or the plant manufacturer.

## 10 Maintenance

Maintenance is to ensure that all functions are maintained or that they can be restored after a breakdown.

Maintenance includes specifications about inspection, service and repairs.

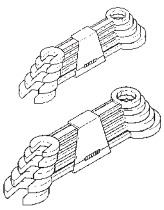
Maintenance includes instructions for trained and qualified personnel.

If anything is unclear consult Aerzener customer service.


During inquiries please state:
- order and serial number
- prevailing faults / malfunctions as accurately as possible
- steps taken to rectify faults.

Is the machine sent back to the supplier, the following measures are to be carried out:
- Completely drain oil, otherwise it is transport of hazardous goods.
- Treat bare components with preservative.
- Seal flange with blind cover.
- Seal open connections.
- Also observe instructions in chapter “Transport”.

### 10.1 Inspections / inspection schedule



A general inspection should be carried out by a service technician from Aerzener after 3 years or 20,000 operating hours.
This includes the preventive maintenance of wearing and replacement parts such as bearings, seal, etc.
We recommend maintaining a stock of replacement and wearing parts to avoid or reduce waiting times and downtimes.

# Maintenance schedule



**Whenever work is carried out on the rotary piston machine, it must be switched off and isolated from the electrical mains supply. Failure to observe this point could result in damage or injury!**
**Various steps should be taken to achieve a long service life and provide the optimum operating conditions. These include carrying out the maintenance work stipulated in this table at the intervals stipulated in this table.**

**We recommend carrying out the maintenance work on the rotary piston machine at the stipulated intervals. The operating hours represent average operating conditions.**
**The actual service life of the equipment may vary depending on environmental conditions and operating data.**
**If you think this situation applies to you, please contact Aerzener Maschinenfabrik.**

**Maintenance times**

| Period in operating hours (Oh) | Mechanics | Lubrication |
|---|---|---|
| **Weekly** | | **Check oil level** |
| ***) After the first 500 Oh** | | **Change oil** |
| ****) after every 4000 Oh,** and at least after 1 year | | **Change oil** |
| **After every subsequent 25,000 Oh** | **- Change wearing parts<br>- Check gearwheels, replace according to state of wear.** | |
| *****) after every 25,000 Oh,** and at least after 3 year | | **Change PFPE oil**<br>under normal, clean operating conditions |

***)** Under normal, clean operating conditions, the basic performance of the lubricant only changes due to the wear on the bearings and gearwheel while running in.
We therefore recommend removing any abrasion residues from the running-in phase by changing the lubricant after the first **500 Oh**.

****) Mineral oils / synthetic oils**
Providing the operating conditions are normal, it is sufficient to change the lubricant every **4000 Oh** or at least **once a year**.
Shorter intervals will be necessary for the conveying of corrosive vapours / gases and/or high dust levels in association with frequent changes between atmospheric and vacuum pressure.
Under these conditions, we recommend determining the state of the oil with regular oil tests.

*****) PFPE oils**
**PFPE lubricants** (e.g.Fomblin) do not wear out if used properly.
Due to its outstanding, chemical, thermal and mechanical stability, its basic performance generally does not change.
Under normal operating conditions therefore, it is possible to achieve service lives above **25000 Oh or 3 years**.
It may be necessary to change the lubricant before this if the oil is contaminated and/or the oil properties change due to the process gas.
Whether it is necessary to change the PFPE may have to be decided by carrying out an oil test.
We therefore recommend changing the PFPE **after one year** at the latest.
Dark discoloration of the lubricant indicates mechanical wear in the bearings, gears and/or seals. The machine must be immediately taken out of service and repaired.

**Arrange for the rotary piston machine to undergo a complete check by the Aerzener Service department at the stipulated intervals**
**and certainly at least once a year.**
**Or: Take out a maintenance contract with Aerzener Maschinenfabrik.**
**Assuming that maintenance work is carried out regularly and professionally,**
**Aerzener Maschinenfabrik is confident that your machine will offer maximum reliability for your application.**

## 10.3 Lubricating oil specifications

Lubricating oils with the following properties must be used for the lubrication of the vacuum rotary blower:

**Lubricating oil for oil chambers**
High ageing resistant fully synthetic oil with good, natural corrosion and ageing protection.

Lubricating oil which meets the minimum requirements as per:
<u>Type CL according to DIN 51 517 Part 2</u>.

At 100°C, the lubricating oil must not exceed a vapour pressure of 1 x 10-4 mbar.

If installed outdoors, make sure that the pour point of the lubricating oil used is below the lowest ambient temperature.

| Lubricating oil manufacturers | Oil types |
|---|---|
| | |
| FRAGOL | Anderol 555** |
| | |
| | |
| **) Ester oil | |
| Table 001 | |

## Lubricating oil when conveying helium

| Lubricating oil manufacturers | Oil types |
|---|---|
| | |
| Cognis | Breox Lubricant B 35 |
| | |

Table 002

"The lubrication recommendations issued by Aerzener Maschinenfabrik relate to the company's own tests and operating experience. Aerzener Maschinenfabrik reserves the right to withdraw the lubrication recommendations if the lubricant or material properties of the machine elements change or if the lubricant is found not to be suitable from practical experience."

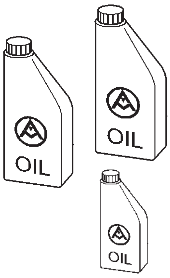
## Lubricating oil for use in the food industry

| Lubricating oil manufacturers | Oil types |
|---|---|
| | |
| SOLVAY SOLEXIS | Fomblin YLVAC 25/6 |
| | |

Table 003

Fomblin oil is approved according to NSF-H1 and NSF-H2.

ENGLISH




## 10.4 Lubricating oil volumes

| Oil filling, in approx. litres / gear side and drive side | | |
|---|---|---|
| **Blower type** | **GL ... Horizontal delivery** | **GM ... Vertical delivery** |
| 10.0 10.1 GM/GL 170 HM / 250 HM | 0,7 | 0,9 |
| 11.3 11.4 GM/GL 500 HM / 700 HM | 0,8 | 1,1 |
| 12.5 12.6 GM/GL 1000 HM / 1400 HM | 1,1 | 1,5 |
| 13.f7 13.8 GM/GL 2000 HM / 2800 HM | 2,1 | 2,5 |
| 14.9 GM/GL 4000 HM | 4,6 | 7,5 |
| 15.10 15.11 GM/GL 5500 HM / 7500 HM | 7,0 | 11,5 |
| 16.12 16.f13 16.13 GM/GL 8000 HM / 9500 HM | 7,6 | 12,0 |
| | | |
| | | |
| | | |
| | | |
| | | |

**NOTE:** The quantities of lubricating oil stated here are guide values. To determine the quantity of oil to be poured in, monitor the indicator in the blower's oil sight glasses.

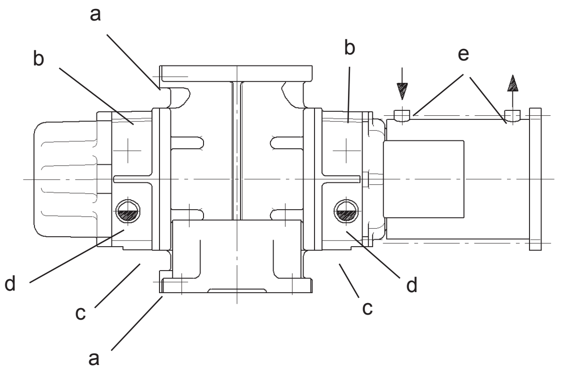
## Filling oil chambers with lubricating oil / GM / GL

- The oil drains (c) must be sealed closed.
  Ensure the valves or sealing plugs are correctly seated!
- Open the oil fillers (b1).
- Fill with oil. Fill to bottom of the sight glasses.
  Allow oil to run down the inside walls.
- After waiting a short time, the oil level will reach a steady state.
- The oil level must now be topped up according to the information stated below.
- Seal the oil fillers (b1) with a new sealing ring.

Make sure that each of the oil chambers is filled and drained separately and that they are checked at the two sight glasses.

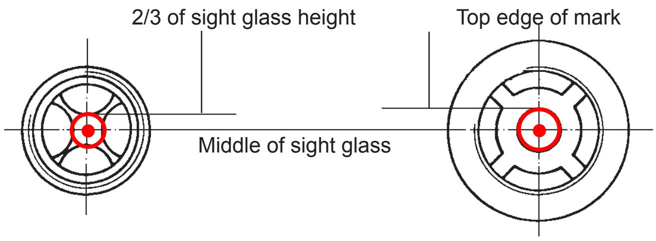


**GM 1000 HM - GM 9500 HM**




a - Instrument connection
b1 - Oil filler, blower
c - Oil drain / drain valve
d - Oil level
e - Coolant connections


## Lubricating oil level according to lubricating oil recommendation




| Blower type | | Lubricating oil level display |
|---|---|---|
| GMa 12 - 13 | GM/GL 1000HM - 2800HM | 1/2 sight glass height |
| GLa 12 - 13 | GM/GL 1000HM - 2800HM | 2/3 sight glass height |
| GMb 14 - 16 | GM/GL 4000HM - 9500HM | 1/2 sight glass height |
| GLb 14 - 16 | GM/GL 4000HM - 9500HM | Top edge of mark |
| | | |
| Filling level with machine stationary | | |

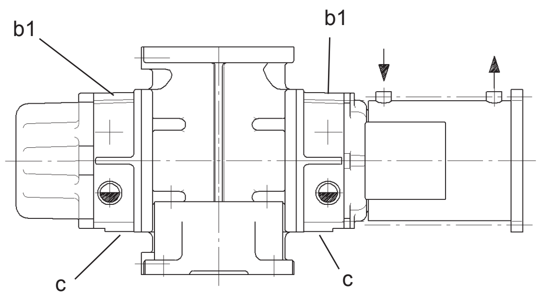
## Lubricating oil level with Fomblin oil filling

| Blower type | | Lubricating oil level display |
|---|---|---|
| GMa 12 - 13 | GM/GL 1000HM - 2800HM | 2/3 sight glass height |
| GLa 12 - 13 | GM/GL 1000HM - 2800HM | 2/3 sight glass height |
| GMb 14 - 16 | GM/GL 4000HM - 9500HM | 2/3 sight glass height |
| GLb 14 - 16 | GM/GL 4000HM - 9500HM | 2/3 sight glass height |
| | | |
| Filling level with machine stationary | | |

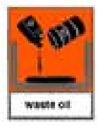
# Draining lubricating oil

## GM 1000 HM - GM 9500 HM

b1 -Oil filler, blower
c - Oil drain / drain valve




- To bleed the oil chamber, open screw plugs b1 and b2, if present.
- Remove screw plugs c.
- Collect the waste oil in a container.
- The waste oil should be disposed of properly and in an environmentally-friendly manner.
- Fit screw plugs correctly with a new sealing ring.
- Check for leaks.

# Error diagnostics/Troubleshooting/Maintenance

**Repairs to rotary piston machines may only be performed by authorised and trained personnel. Improper repairs may result in considerable risks to the user/ personnel.**

| Error messages/ Faults What do I do if . . ? | Possible causes . . | Remedy . . |
|---|---|---|
| ...abnormal running noises occur? | • Damaged bearings<br>• Rotary pistons coming into contact with one another or with the conveying chamber<br>• Rotary pistons coming into contact due to contamination<br>• Foreign bodies in gear wheels<br>• Shaft deflection | • Replace bearings<br>• Check play settings, check cylinder for cracks<br>• Clean conveying chamber<br>• Check, clean and, if necessary, replace gear wheels<br>• Measure shaft deflection, replace rotary pistons if necessary |
| ...the blower becomes too hot? | • Intake filter soiled<br>• Ambient temperature too high<br>• Hood inlet openings blocked<br>• Hood fan faulty<br>• Oil level or oil viscosity too high<br>• Rotary piston play too great<br>• Overloaded | • Replace filter<br>• Ensure that there is adequate ventilation<br>• Clean acoustic hood air supply<br>• Replace fan<br>• Correct oil level, note the viscosity<br>• Replace damaged components<br>• Check and comply with operating data |
| ... is there oil in the material to be conveyed? | • Oil level too high<br>• Worn seal | • Correct oil level<br>• Replace seals |
| ... is the intake volume too low? | • Starting strainer/intake filter contaminated<br>• Lines leaking<br>• Blower incorrect dimensions<br>• Damaged rotary pistons / cylinder | • Clean, replace if necessary<br>• Seal lines<br>• Check design<br>• Replace damaged components |
| | | |

| Error messages/ Faults What do I do if . . ? | Possible causes . . | Remedy . . |
|---|---|---|
| ... does the motor draw too much power? | • Operating data does not agree with ordering data<br>• Mechanical damage to blower or motor<br>• Drop in motor voltage | • Check and comply with operating data<br>• Replace damaged components<br>• Adjust power; see motor instructions |
| ... does the blower rotatation reverse after shutdown? | • Non-return valve faulty or not tight | • Replace valve |
| ... if the machine is damaged, is this because of a faulty mains connection? | • Motor and electr. control have been connected to two different mains power supplies. | • Connect motor and control voltage to a single mains power supply.<br>• Alternatively: Use a current monitoring relay. |
| Whenever it is necessary to intervene because of a fault, the following must be checked:<br>• Freedom of running of the rotary piston machine<br>• Contact-free rotation<br>• Lubricating oil level<br>• Correct operation and connection<br>• Compliance with the safety and warning information | | |
| | | |

# Spare parts - overview

For effective and durable use of the rotary piston machine, there follows a list of components necessary for maintenance, an inspection or a repair.
The components are grouped by expected life into three spare part versions.



**1.) Maintenance**

- Shaft sealing ring(s)
- Inner ring, shaft sleeve, bush
- Sealing rings and screw plugs, drain valves for oil drain
- V-belt / wearing parts on coupling
- Air filter element
- Lubricating oil, oil filter element

**2.) Inspection**

- Components from point 1.)
- Bearing and sealing components
- for compressor plus the adjustable bearing cover including the related minor items

**3.) Repair**

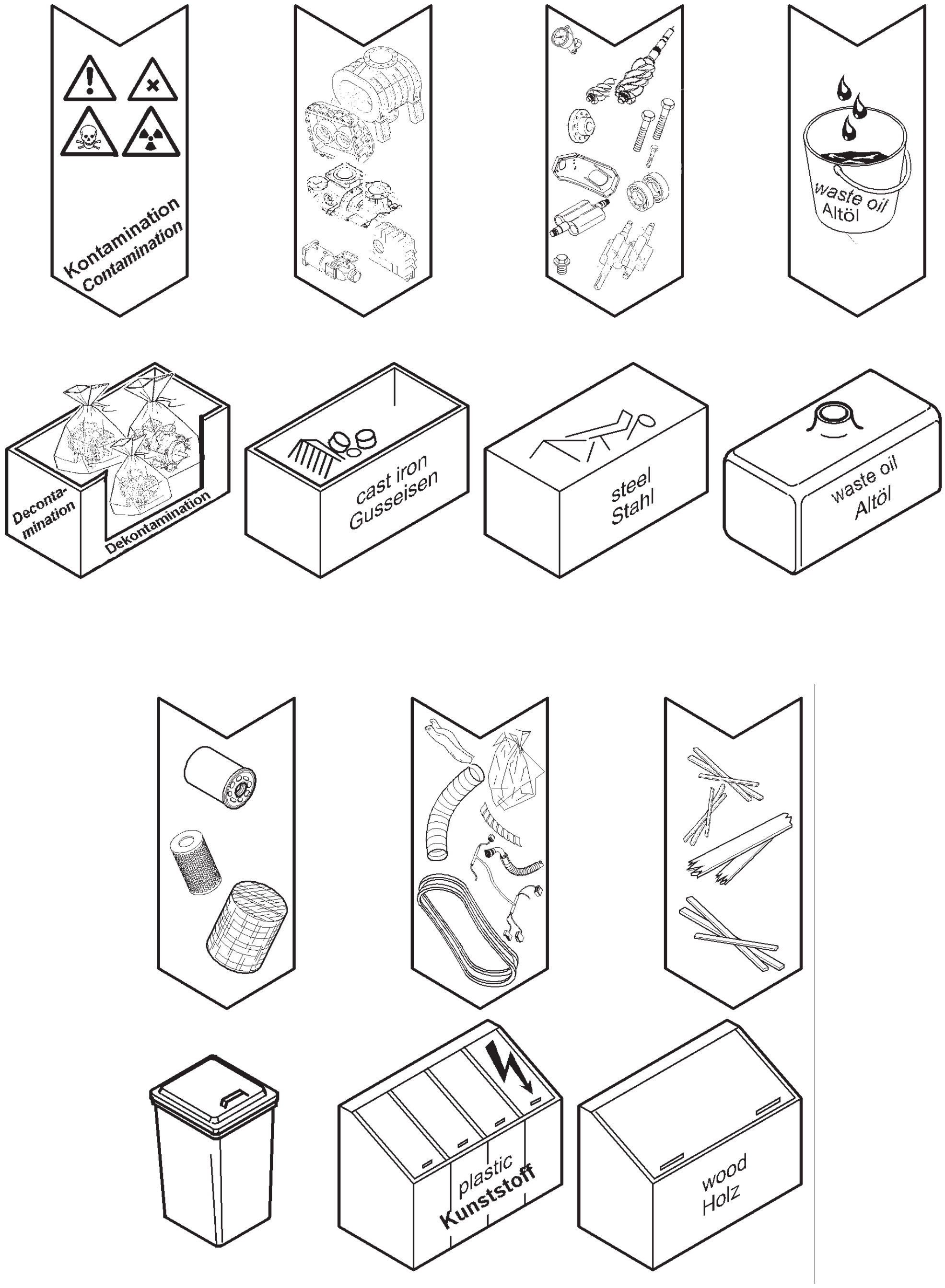
- Components from point 1.) and 2.)
- Rotor pair / rotary piston pair
- Timing gear wheels
- for compressor plus the torsion shaft and the oil pump

**Spare parts and accessories**

It is specifically highlighted that original parts and accessories not supplied by us have also not been tested or approved by us.
Therefore, installing or mounting and using any such products could, under certain circumstances, degrade the stipulated system characteristics. The manufacturer shall assume no liability whatsoever for damage caused by the use of non-genuine parts or accessories.

## Recycling / Disposal

- All waste products are to be disposed of or treated not harmful to the environment.
- Used lubricants are to be disposed of properly.
- Contaminated components and auxiliary material are to be packed and decontaminated.



# INFO - SEITE

*Information sheet* *Page infos*
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# **Anhang I/** *Appendix I*

**elektrische Motordaten**/ *electrical motor data*

| | Motortyp motortype | Gebläsetypen blower type | Spannung voltage | Frequenz frequency | Leistung power | | Eckspannung voltage set point | Eckfrequenz frequency set point | Kurzzeitig short term operation < 5 Minuten | Dauerbetrieb Continuous operation cool down |
|---|---|---|---|---|---|---|---|---|---|---|
| | elektrische Motordaten electrical motor data IP 65 | | | | | | Parameter Frequenzumrichter Set of parameters variabale frequency drive | | Mode S6 I nenn current | Mode S1 I nenn current |
| | | GM / GL HV | Stern / Dreieck star / delta | | Mode S6 overload < 5 min | Mode S1 continuous operation / cool down | Stern / Dreieck star / delta | | Stern / Dreieck star / delta | Stern / Dreieck star / delta |
| | | | V | Hz | kW | kW | V | Hz | A | A |
| | HDM 150 / 120 | 1000HM-Y/ 1400HM-Y | 70- 400 / 40-230 | 20 - 120 | 15,0 | 1,0 | 400 / 230 | 100 | 29,5 / 51,1 | 5,6 / 9,6 |
| | HDM 150 / 120 | 2000HM-C/ 2800HM-C | 60-400 / 35-230 | 16 - 100 | - | 9,0 | 400/ 230 | 100 | - | 18,7/ 32,4 |
| **EM*** | HDM 200 / 200 | 2000HM-Y/ 2800HM-Y | 65-400 / 37-230 | 16 - 100 | 20,0 | 3,0 | 400 / 230 | 100 | 41*/ 71* | 16/ 28 |
| **PM*** | HDM 200 / 200 | 2000HM-Y/ 2800HM-Y | 65-400 / 37-230 | 16 - 100 | 30,0 | 3,0 | 400 / 230 | 100 | 60/ 104 | 16/ 28 |
| | HDM 200 / 200 | 4000HM-C | 65-400 / 37-230 | 16 - 100 | - | 20,0 | 400 / 230 | 100 | - | 41 / 71 |
| | HDM 200 / 200 | 5500HM-C/ 7500HM-C | 80 -290/ 45-167 | 16 - 60 | - | 15,0 | 245/ 142 | 60 | - | 47 / 81 |
| **EM*** | HDM 240/ 200 | GM 4000HM-Y | 50-400 / 29-230 | 16 - 80 | 29 | 12,0 | 400/ 230 | 60 | 59* / 102* | 30/52 |
| **PM*** | HDM 240/ 200 | GM 4000HM-Y | 50-400 / 29-230 | 16 - 80 | 35 | 12,0 | 400/ 230 | 60 | 70,5/ 122 | 30/52 |
| | HDM 240/ 200 | 5500HM-N/ 7500HM-N | 400 / 230 | 50 | - | 18,5 | 400 / 230 | 50 | - | 34,5 / 60 |
| | HDM 240/ 200 | 5500HM-N/ 7500HM-N | 460 / 265 | 60 | - | 22,0 | 460 / 265 | 60 | - | 35,5 / 61,5 |
| **EM*** | HDM 240/ 200 | 5500HM-Y/ 7500HM-Y | 110 - 400 / 62 - 230 | 16 - 60 (FU) | 23,0 | 4,0 | 400 / 230 | 60 | 50/86.6* | 24 / 42 |
| **PM*** | HDM 240/ 200 | 5500HM-Y/ 7500HM-Y | 110 - 400 / 62 - 230 | 16 - 60 (FU) | 40,0 | 4,0 | 400 / 230 | 60 | 82/ 142 | 24 / 42 |
| | HDM 240/ 200 | 8000HM-C/ 9500HM-C | ıuf Anfrage/ *on request* | | | | | | | |
| **EM*** | HDM 240 / 300 | 8000HM-Y/ 9500HM-Y | 160-400/ 92-230 | 16 - 40 | 36 | 4,0 | 400 / 230 | 40 | 67 /116* | 23 / 40 |
| **PM*** | HDM 240 / 300 | 8000HM-Y/ 9500HM-Y | 160-400/ 92-230 | 16 - 40 | 53,1 | 4,0 | 400 / 230 | 40 | 105 / 182 | 23 / 40 |

***EM = Efficiency mode:** Verringerte Motorleistung, erreichbare Druckdifferenz siehe Anhang II
****EM = Efficiency mode:*** *Reduced motor rating, available differential pressure, please refer appendix II*
***PM = Performance mode:** Vollständige Motorleistung, erreichbare Druckdifferenz siehe Anhang II.
****PM = Performance mode:*** *full motor rating, available differential pressure, please refer appendix II*
**ACHTUNG**: Bei Nichtbeachtung der gebläsespezifischen K0-/ bzw. Betriebssicherheitskurven besteht die Gefahr einer Überhitzung des Gebläses
**ATTENTION:** In case of non-observance of the blower-specific K0-/ resp. operation safety curves, there is the risk of blower overheating!

*) Werte können von den Angaben auf dem Motortypenschild abweichen, die Stromaufnahme des Motors hängt von der zulässigen Druckdifferenz des Gebläses ab (siehe auch Anhang II)!
**Beachte:** Für die Auswahl des Frequenzumrichters sind die Angaben dieser Betriebsanleitung (s.o.) ausschlaggebend

# **Anhang II** / *Appendix II*

**Betriebsdaten** / *Operating data*

| Gebläsetypen *blower type* GM / GL HV | Gebläsetypen *blower type* ... CM | Motortyp *motortype* | Kurzzeitbetrieb von 5 Minuten, S6 / *short-term operation of 5 minutes, S6* max. delta p (mbar) | Dauerbetrieb Nennbereich, S1 / *continous operation nominal range, S1* max. delta p (mbar) | min. Drehzahl / *min. speed* Hz *c/s* | max. Drehzahl / *max. speed* Mineral-Öl / oil Hz *c/s* | max. Drehzahl / *max. speed* Fomblin Öl / oil Hz *c/s* | max. Wicklungs-temperatur / *max. winding temperature* °C | Kühlmittelbedarf / *coolant consumption* (l/min) delta t= 10K | Kühlmittel Nachlaufzeit / *coolant afterrunning time* (min) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 12.5 | 1000 HM-Y | HDM 150/120 | 150 | 50 | 20 | 120 | 120 | 150 | 3 | 5 | 120 Hz Zyklusbetrieb cycle operation |
| 12.6 | 1400 HM-Y | HDM 150/120 | 100 | 30 | 20 | 120 | 120 | 150 | 3 | 5 | |
| 13.f7 | 2000 HM-C | HDM 150/120 | / | 50 | 16 | 100 | 100 | 150 | 3 | 5 | 100 Hz Dauerbetrieb continous operation |
| 13.8 | 2800 HM-C | HDM 150/120 | / | 30 | 16 | 100 | 100 | 150 | 3 | 5 | |
| 13.f7 | 2000 HM-Y | HDM 200/200 | 150 EM* | 50 | 16 | 100 | 100 | 150 | 4 | 5 | 100 Hz Zyklusbetrieb *cycle operation* |
| 13.f7 | 2000 HM-Y | HDM 200/200 | 230 PM* | 50 | 16 | 100 | 100 | 150 | 4 | 5 | |
| 13.8 | 2800 HM-Y | HDM 200/200 | 100 EM* | 30 | 16 | 100 | 100 | 150 | 4 | 5 | |
| 13.8 | 2800 HM-Y | HDM 200/200 | 160 PM* | 30 | 16 | 100 | 100 | 150 | 4 | 5 | |
| 14.9 | 4000 HM-C | HDM 200/200 | / | 70 | 16 | 100 | 100 | 150 | 4 | 5 | 100 Hz Dauerbetrieb continous operation |
| 14.9 | 4000 HM-Y | HDM 240/200 | 150 EM* | 50 | / | / | / | / | / | / | / |
| 14.9 | 4000 HM-Y | HDM 240/200 | 185 PM* | 50 | / | / | / | / | / | / | / |
| 15.10 | 5500 HM-C | HDM 200/200 | / | 65 | 16 | 60 | 60 | 150 | 4 | 5 | 60 Hz Dauerbetrieb continous operation |
| 15.11 | 7500 HM-C | HDM 200/200 | / | 45 | 16 | 60 | 60 | 150 | 4 | 5 | |
| 15.10 | 5500 HM-Y | HDM 240/200 | 110 EM* | 70 | 16 | 60 | 60 | 150 | 4 | 5 | 60 Hz Zyklusbetrieb *cycle operation* |
| 15.10 | 5500 HM-Y | HDM 240/200 | 200 PM* | 70 | | | | | | | |
| 15.11 | 7500 HM-Y | HDM 240/200 | 80 EM* | 55 | 16 | 60 | 60 | 150 | 4 | 5 | |
| 15.11 | 7500 HM-Y | HDM 240/200 | 150 PM* | 55 | | | | | | | |
| 15.10 | 5500 HM-N | HDM 240/200 | / | 100 | 16 | 50/60 | 50/60 | 150 | 4 | 5 | 50/60 Hz Netzbetrieb *net operation* |
| 15.11 | 7500 HM-N | HDM 240/200 | / | 75 | 16 | 50/60 | 50/60 | 150 | 4 | 5 | |
| 16.f13 | 8000 HM-C/Y | HDM 240/300 | 90 EM* | 50 | 16 | 40 | 40 | 150 | 7 | 5 | 40 Hz / 4-polig Zyklusbetrieb *cycle operation* |
| 16.f13 | 8000 HM-C/Y | HDM 240/300 | 130 PM* | 50 | 16 | 40 | 40 | 150 | 7 | 5 | |
| 16.13 | 9500 HM-C/Y | HDM 240/300 | 75 EM* | 45 | 16 | 40 | 40 | 150 | 7 | 5 | 40 Hz / 4-polig Zyklusbetrieb *cycle operation* |
| 16.13 | 9500 HM-C/Y | HDM 240/300 | 110 PM* | 45 | 16 | 40 | 40 | 150 | 7 | 5 | |

*EM = Efficiency mode / reduzierte Motorleistung, siehe Anhang I / reduced motor rating, please refer appendix I

*PM = Performance mode / vollständige Motorleistung, siehe Anhang I / full motor rating, siehe appendix I

Datum: 28-10-2010

## Anhang III
**Trudeldrehzahl HV-Gebläse**

<table>
<tr><td></td><td>max. zul. Drehzahlen für Trudelbetrieb mit FU (1/min)</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="3"></td></tr>
<tr><td>Typ / type</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>GM/GL170 HM</td><td>190</td><td colspan="9" rowspan="15">Bei Trudelbetrieb ist sicherzustellen, dass spätestens alle 24 Stunden ein Hochlauf auf die Nenndrehzahl erfolgt, um eine hinreichende Ölversorgung von Lagern und Zahnrädern sicherzustellen.<br><br>Erfolgt dieser Hochlauf nicht, sind Schäden an diesen Bauteilen nicht auszuschließen!</td></tr>
<tr><td>GM/GL 250 HM</td><td>190</td></tr>
<tr><td>GM/GL 340 HM</td><td>190</td></tr>
<tr><td>GM/GL 500 HM</td><td>150</td></tr>
<tr><td>GM/GL 700 HM</td><td>150</td></tr>
<tr><td>GM/GL 1000 HM</td><td>120</td></tr>
<tr><td>GM/GL 1400 HM</td><td>120</td></tr>
<tr><td>GM/GL 2000 HM</td><td>90</td></tr>
<tr><td>GM/GL 2800 HM</td><td>90</td></tr>
<tr><td>GM/GL 4000 HM</td><td>75</td></tr>
<tr><td>GM/GL 5500 HM</td><td>60</td></tr>
<tr><td>GM/GL 7500 HM</td><td>60</td></tr>
<tr><td>GM/GL 8000 HM</td><td>50</td></tr>
<tr><td>GM/GL 9500 HM</td><td>50</td></tr>
<tr><td>GM/GL 9500 HM</td><td>50</td></tr>
<tr><td colspan="11">gültig für Betriebsanleitung G4-014 ...<br><br>09-2012</td></tr>
</table>

## Appendix III
**Coasting speed HV blower**

<table>
<tr><td></td><td>Max. perm. speeds for coasting mode with frequency converter (1/min)</td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="3"></td></tr>
<tr><td>Typ / type</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>GM/GL170 HM</td><td>190</td><td colspan="9" rowspan="15">In the case of the coasting mode it is to be ensured that the unit is run up to the rated speed at the latest every 24 hours to ensure an adequate supply of oil to the bearings and gear wheels.<br><br>If the unit is not run up, damage to these parts cannot be excluded!</td></tr>
<tr><td>GM/GL 250 HM</td><td>190</td></tr>
<tr><td>GM/GL 340 HM</td><td>190</td></tr>
<tr><td>GM/GL 500 HM</td><td>150</td></tr>
<tr><td>GM/GL 700 HM</td><td>150</td></tr>
<tr><td>GM/GL 1000 HM</td><td>120</td></tr>
<tr><td>GM/GL 1400 HM</td><td>120</td></tr>
<tr><td>GM/GL 2000 HM</td><td>90</td></tr>
<tr><td>GM/GL 2800 HM</td><td>90</td></tr>
<tr><td>GM/GL 4000 HM</td><td>75</td></tr>
<tr><td>GM/GL 5500 HM</td><td>60</td></tr>
<tr><td>GM/GL 7500 HM</td><td>60</td></tr>
<tr><td>GM/GL 8000 HM</td><td>50</td></tr>
<tr><td>GM/GL 9500 HM</td><td>50</td></tr>
<tr><td>GM/GL 9500 HM</td><td>50</td></tr>
<tr><td colspan="11">applies to operating instructions G4-014 ...<br><br>09-2012</td></tr>
</table>

Technische Dokumentation


Technical Documentation


Aerzener Maschinenfabrik GmbH

| | |
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